National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Microprocessor Interfacing and Programming	Course Code:	EE3002
Program:	BS Electrical Engineering	Semester:	Fall 2025
Duration:	15 minutes	Total Marks:	10
Quiz Date:	23-Oct 2025	Weight:	2.5
Section:	BEE-5C	Page(s):	1
Exam Type:	Quiz- 2	CLO #	2

Student Name: [redacted]

Instructions:
1. Do not forget to write your Name and Roll Numbers.
2. Solve on the paper and Return.

Question No. 1 (CLO No. 2) Marks:

Two temperature sensors are connected with PIC microcontroller at RB0 and RB1. The values received at RB0 and RB1 are 68 and 74, respectively. Construct the PIC Assembly program to find the sum of both numbers. Sum must be in BCD, and saved in 20H.

```
TEMP1   EQU  0x30  ; 68
TEMP2   EQU  0x31  ; 74
STOR1   EQU  0x20

BSF     PORTB,0   ; input RB0
BSF     PORTB,1   ; input RB1

; Assum recieved 68 in TEMP2, 74 in TEMP2 now.

MOVF    TEMP1,W   ; move Temp1 to wreg.
ADDWF   TEMP2,W   ; Add Temp2 to WREG, save result in WREG
DAW               ; Adjust for BCD.
MOVWF   STOR1
END
```

Question No. 2 (CLO No. 2) EP **Marks:**

Construct the PIC Assembly program to read the values from sensors connected at RD7 and RD6, and save the values in RC0 and RC7. Before saving the read numbers, ensure that the value 7 with carry is added in each of them.

```
BSF   PORTD,7    ; input
BSF   PORTD,6    ; input

~~START EQU 0~~

MOVF  PORTD,7
ADDLWC d'7'
MOVWF PORTC,0

MOVF  PORTD,6
ADDLWC d'7'
~~MOVWF PORTD,6~~
MOVWF PORTC,6
END.
```



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